

6.4 Work

What is work?

Definition:

_____ is any influence that causes an object to undergo a change in speed, direction, or shape. _____ has magnitude and direction. We say that an object with mass (m) and acceleration (a) has force

*Recall that the gravitational constant is 9.8 m/s^2 .

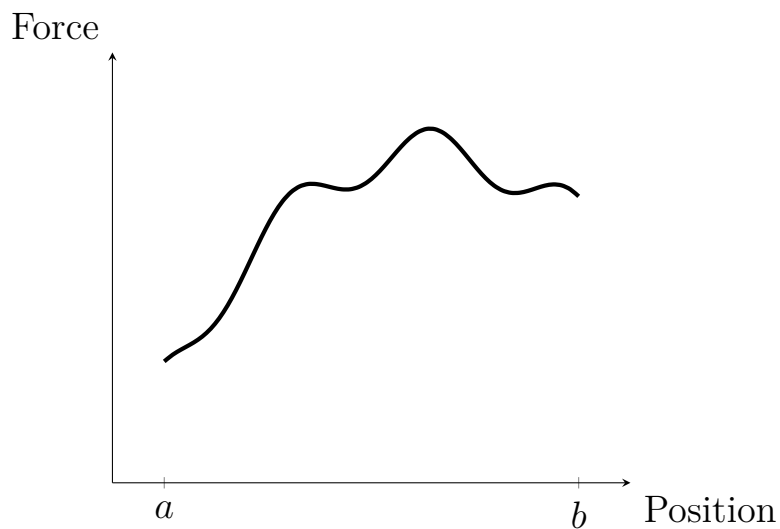
A body moves a _____, d , along a straight line after being acted by a _____ of constant magnitude, F , in the direction of motion, we define the _____, W , done by the force on the body with the formula

	Force F	distance d	Work W
units (US/Imperial System)			
units (SI/International System)			

Example 1

Determine the work done by a weightlifter in raising a 60 kg barbell from the floor to a height of 2 meters.

Suppose the amount of force is not constant, but rather depends on the position of the object.



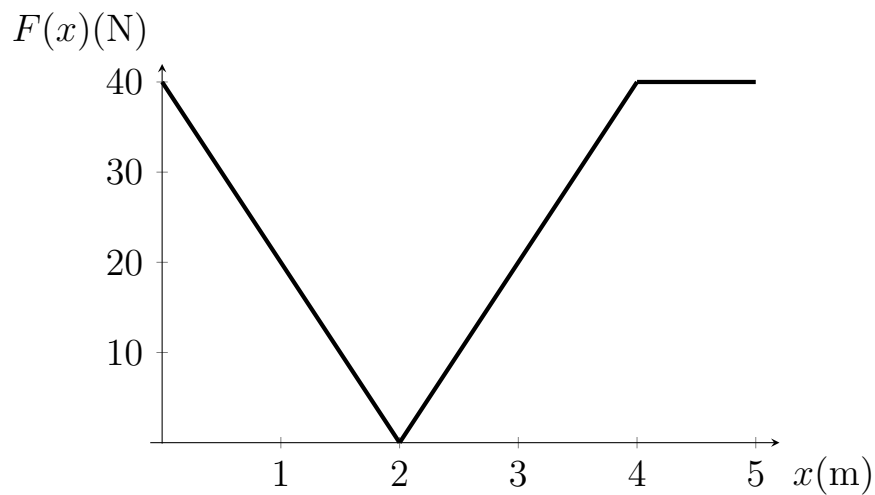
Definition:

The _____ done by a variable force, $F(x)$, in the direction of motion along the x -axis from $x = a$ to $x = b$ is

$$W =$$

Example 2

The graph of a force function (in Newtons) is shown below. How much work is done by the force in moving an object a distance of 5 m?



Work Done When Compressing or Stretching a Spring

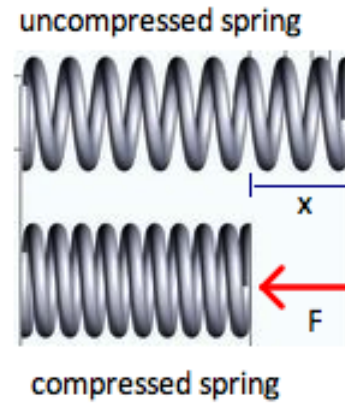
Hooke's Law

The force required to compress or stretch a spring by x units short of or beyond its natural length is proportional to x

$$F =$$

Note that k is called the

_____.



Example 3

A spring has a natural length of 50 cm. A force of 50N stretches the spring to a length of 60 cm.

(a) What force is needed to stretch the spring x meters ?

(b) How long is the spring when stretched by a force of 200 N?

- (c) Set up an integral to find the work done in stretching the spring 20 cm.
- (d) Set up an integral to find the work done in stretching the spring from a length of 100 cm to 110 cm.
- (e) 20 J of work is done in stretching the spring a certain distance.
How far was the spring stretched?

Summary of Springs

- The variable of integration represents:
- How do we find the bounds for our integral?
- Why do we not use a negative sign to indicate a spring is being compressed?

Pulley Systems

In pulley systems the weight that is needed to be lifted depends on how much rope is drawn. This is another example of a system where the force depends on position.

Example 4

A uniform cable 30 ft long and weighing 60 lbs hangs vertically from a pulley system at the top of a building. A steel beam weighs 500 lbs and is attached to the end of the cable.

- (a) Set up an integral to find the work required to pull the beam to the top.



(b) Set up an integral to find the work required to pull the beam up 10 ft.